LAPORAN PRAKTIKUM

POSTTEST 1

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

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Kelas C2 ‘25

PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

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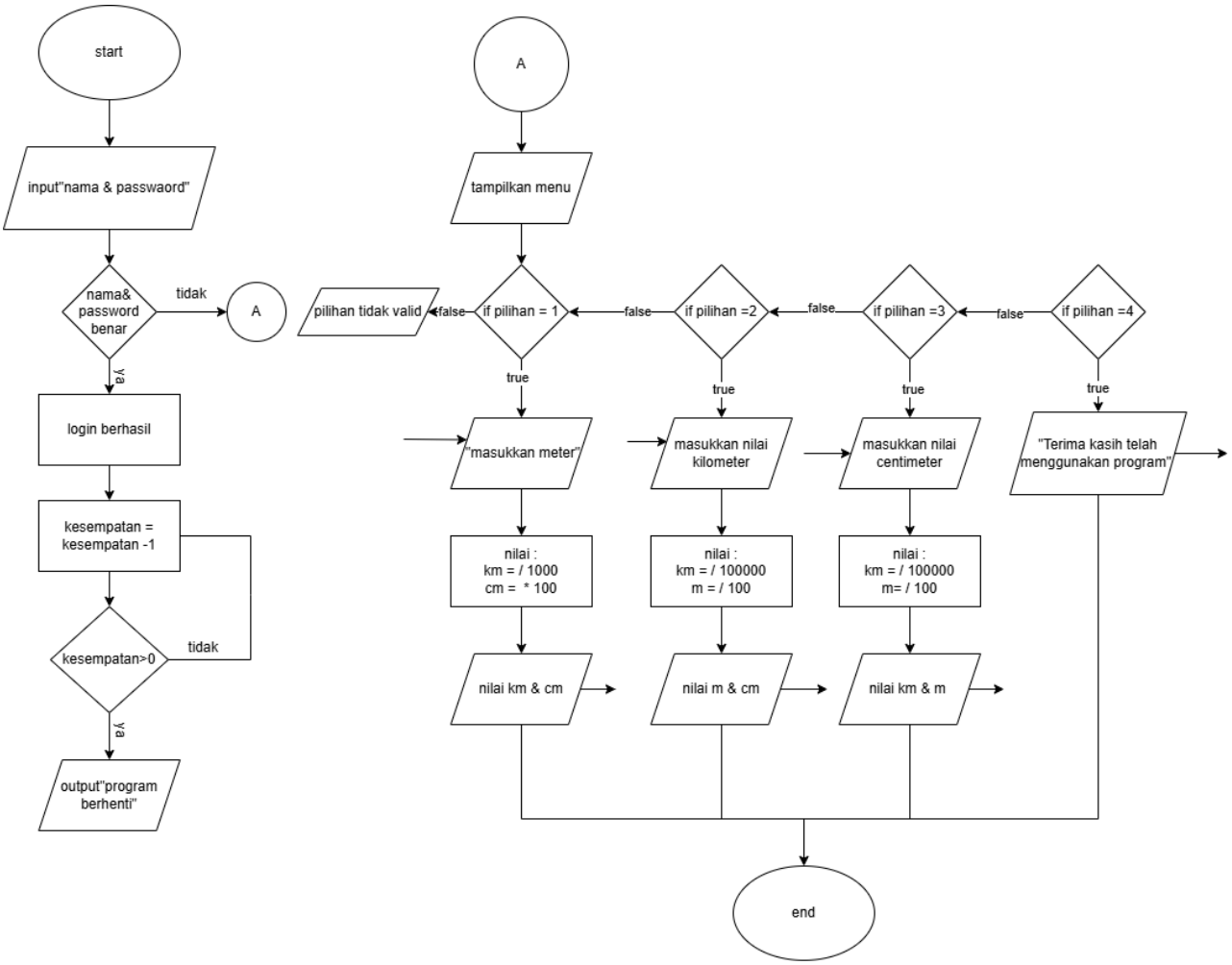
1. Flowchart


start
input"nama & passwaord"
nama& password benar
tidak
A
ya
login berhasil
kesempatan = kesempatan -1
kesempatan>0
tidak
ya
output"program berhenti"
A
tampilkan menu
pilihan tidak valid
false
if pilihan = 1
false
if pilihan =2
false
if pilihan =3
false
if pilihan =4
true
true
true
true
"masukkan meter"
masukkan nilai kilometer
masukkan nilai centimeter
"Terima kasih telah menggunakan program"
nilai : km = / 1000 cm = * 100
nilai : km = / 100000 m = / 100
nilai : km = / 100000 m= / 100
nilai km & cm
nilai m & cm
nilai km & m
end

2. Deskripsi Singkat Program

Tujuan

- *Login sistem sederhana: pengguna harus memasukkan nama dan password yang benar, dengan batas maksimal 3 kali percobaan. Ini melatih penggunaan loop, if-else, dan validasi input.*

- *Menu interaktif: setelah login berhasil, program menampilkan menu konversi satuan panjang (meter, kilometer, centimeter). Ini melatih penggunaan switch-case dan do-while loop.*

Fungsinya :

- *Memberikan pengalaman membuat program dengan autentikasi sederhana.*
- *Menyediakan konversi satuan panjang secara otomatis, sehingga pengguna tidak perlu menghitung manual.*
- *Melatih keterampilan dasar C++ seperti input/output, perulangan, percabangan, dan format angka dengan iomanip.*

3. Source Code

Source Code:

```
#include <iostream>
#include <iomanip>
#include <string>
using namespace std;

int main() {
    string nama, password;
    string namaBenar = "Maynard";
    string passBenar = "114";   // 3 digit terakhir NIM
    int kesempatan = 3;

    // ================= LOGIN =================
    while (kesempatan > 0) {
        cout << "====================================" << endl;
        cout << "            LOGIN PROGRAM           " << endl;
        cout << "====================================" << endl;
```

```
        cout << "Masukkan Nama     : ";
        cin >> nama;
        cout << "Masukkan Password : ";
        cin >> password;

        if (nama == namaBenar && password == passBenar) {
            cout << endl;
            cout << "Login Berhasil! Selamat datang, " << nama << endl;
            break;
          }
            else {
            kesempatan--;
            cout << endl;
            cout << "Login gagal! Sisa kesempatan: " << kesempatan <<
endl;
            cout << "-------------------------------------" << endl;
        }
    }

    if (kesempatan == 0) {
        cout << endl;
        cout << "Anda gagal login 3 kali. Program berhenti." << endl;
        return 0;
    }

    // ================= MENU =================
    int pilihan;
    double nilai;

    do {
        cout << endl;
        cout << "=====================================" << endl;
        cout << "        MENU KONVERSI PANJANG       " << endl;
        cout << "=====================================" << endl;
        cout << "1. Meter -> Kilometer & Centimeter" << endl;
        cout << "2. Kilometer -> Meter & Centimeter" << endl;
        cout << "3. Centimeter -> Meter & Kilometer" << endl;
        cout << "4. Keluar" << endl;
        cout << "-------------------------------------" << endl;
        cout << "Pilih menu (1-4): ";
        cin >> pilihan;

        switch (pilihan) {
            case 1:
                cout << endl;
```

```
                cout << "Masukkan nilai dalam Meter: ";
                cin >> nilai;
                cout << fixed << setprecision(2);
                cout << "Hasil:" << endl;
                cout << nilai / 1000 << " km" << endl;
                cout << nilai * 100 << " cm" << endl;
                break;

            case 2:
                cout << endl;
                cout << "Masukkan nilai dalam Kilometer: ";
                cin >> nilai;
                cout << fixed << setprecision(2);
                cout << "Hasil:" << endl;
                cout << nilai * 1000 << " m" << endl;
                cout << nilai * 100000 << " cm" << endl;
                break;

            case 3:
                cout << endl;
                cout << "Masukkan nilai dalam Centimeter: ";
                cin >> nilai;
                cout << fixed << setprecision(2);
                cout << "Hasil:" << endl;
                cout << nilai / 100 << " m" << endl;
                cout << nilai / 100000 << " km" << endl;
                break;

            case 4:
                cout << endl;
                cout << "Terima kasih telah menggunakan program" <<
endl;
                break;

            default:
                cout << endl;
                cout << "Pilihan tidak valid!" << endl;
        }

    } while (pilihan != 4);

    return 0;
}
```

4. Hasil Output

```
=====================================
          LOGIN PROGRAM
=====================================
Masukkan Nama     : Maynard
Masukkan Password : 114

Login Berhasil! Selamat datang, Maynard

=====================================
         MENU KONVERSI PANJANG
=====================================
1. Meter → Kilometer & Centimeter
2. Kilometer → Meter & Centimeter
3. Centimeter → Meter & Kilometer
4. Keluar
-------------------------------------
Pilih menu (1-4): 10000

Pilihan tidak valid!
```

Gambar 4.1

```
=====================================
         MENU KONVERSI PANJANG
=====================================
1. Meter → Kilometer & Centimeter
2. Kilometer → Meter & Centimeter
3. Centimeter → Meter & Kilometer
4. Keluar
-------------------------------------
Pilih menu (1-4): 1

Masukkan nilai dalam Meter: 1000
Hasil:
1.00 km
100000.00 cm
```

Gambar 4.2

```
=====================================
        MENU KONVERSI PANJANG
=====================================
1. Meter → Kilometer & Centimeter
2. Kilometer → Meter & Centimeter
3. Centimeter → Meter & Kilometer
4. Keluar
-------------------------------------
Pilih menu (1-4): 2

Masukkan nilai dalam Kilometer: 1000
Hasil:
1000000.00 m
100000000.00 cm

=====================================
        MENU KONVERSI PANJANG
=====================================
1. Meter → Kilometer & Centimeter
2. Kilometer → Meter & Centimeter
3. Centimeter → Meter & Kilometer
4. Keluar
-------------------------------------
Pilih menu (1-4): 3

Masukkan nilai dalam Centimeter: 100
Hasil:
1.00 m
0.00 km
```

Gambar 4.3

5. Langkah-langkah GIT

5.1 GIT Add

```
PS C:\Users\mayna\Documents\APL\POST- TEST\post-test-apl-1> git init
Initialized empty Git repository in C:/Users/mayna/Documents/APL/POST- TEST/post-test-apl-1/.git/
```

5.2 GIT Commit

```
PS C:\Users\mayna\Documents\APL\POST- TEST\post-test-apl-1> git commit -m "pt-1"
[main (root-commit) 236101c] pt-1
 1 file changed, 102 insertions(+)
 create mode 100644 2509106114-Maynard Christian Tallu Lembang.cpp
```

5.3 GIT Push

```
PS C:\Users\mayna\Documents\APL\POST- TEST\post-test-apl-1> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 996 bytes | 199.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
```